2 Digital literacy: Staying safe online

You will learn

- how to recognise the risks and dangers on the internet
- how to avoid risks and dangers on the internet
- how to use internet content responsibly.



In Student Book 5 you learned what it means to be a responsible internet user. Not everyone who uses the internet behaves responsibly. There are some people who make the internet a dangerous place. There are criminals who attempt to steal from you. Others try to install malicious software on your computer. There are bullies who use the internet to frighten and threaten people.

In this unit you will learn about the risks that the internet brings into our lives. You will learn the steps you need to take to guard against these risks.

Unplugged

Answer these two questions individually. How much do you agree with each statement? Choose the answer that is true for you.

"I feel safe when I am online."	"I am confident that I know how to stay safe online."
1 I feel very safe.	**1** I am very confident.
2 I feel quite safe.	2 I am quite confident.
3 I feel neither safe nor unsafe.	3 I am neither confident nor unconfident.
4 I sometimes worry I am at risk.	4 I don't know very much about staying safe.
5 I think the internet is a dangerous place.	5 I know nothing about staying safe online.

Collect the responses of everyone in the class. Create a graph to present your class's responses.

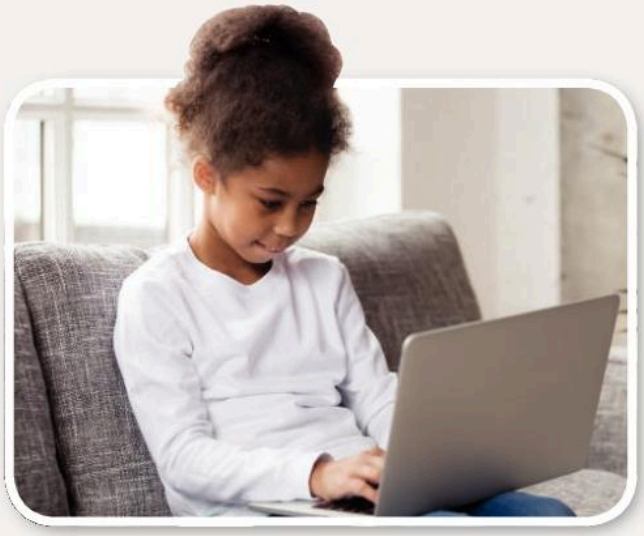
Learning outcomes: Use content from online sources responsibly; Explain risks associated with internet use; Discuss how data may be collected when working online

Cybercrime

Identity theft

Hacking

Malware



Virus

Cyberbully

Ransomware

Talk about...

Some of the threats you face when using the internet are shown above. You will learn more about these threats in this unit, but you may already have heard about some of them. Share what you already know about these threats with your group.

Which of the threats do you worry about most? What do you do to stay safe online?

Did you know?

By 2021 the cost of internet crime is expected to be $6 trillion a year. Businesses across the world will spend $1 trillion to protect themselves against criminals who use the internet.



cybercrime
hacking malware cyberbully
anti-virus software internet shopping
e-commerce secure site cookie
intellectual property copyright
plagiarism firewall

2.1 Collecting data online

In this lesson

You will learn:

- how data is collected online
- how cookies are used on websites.

Finding information online

There are many reasons to look for information. You might do a web search to find information for a geography project. You might look for reviews online to help you decide which model of smartphone is best for you.

In the modern world, the information you need is usually on the internet. When you use the internet you also give away information. In this lesson you will learn about the ways in which websites collect data about you.

Online registration

Websites such as social media and internet shopping sites usually ask you to register before you can use them. **Registration** means becoming a member of a website. You fill in an online form and provide information about yourself to the owner of the site.

In return for registering, you receive some benefits. For example, you might be able to:

- read pages that are hidden from non-registered users
- leave messages in chat rooms and on message boards
- add content
- get email updates when new content is added to the site
- download software.



Which information should you give?

The owner of a website should only collect the information they need, such as a username and password. They may also need contact details such as an email address. Some websites ask for optional information such as your telephone number or birthday. You don't have to give optional information to register. Always think about what information you need to provide when you complete forms online.

Activity

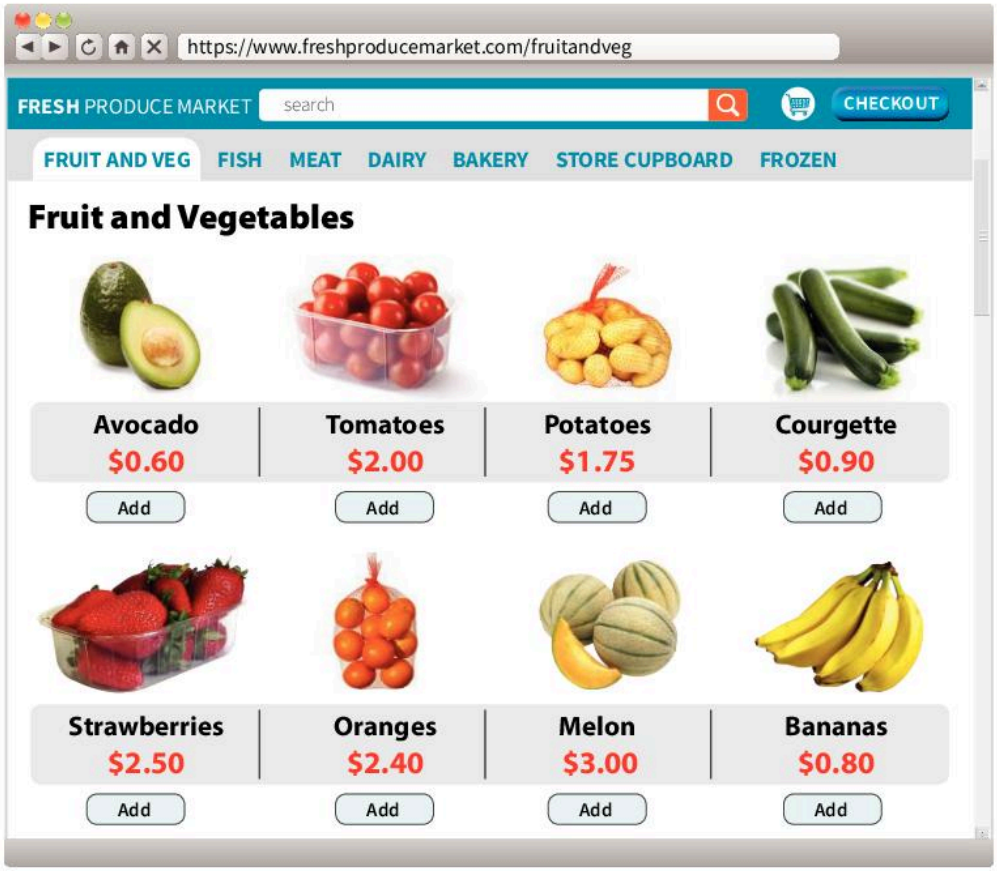
Imagine you have designed a new website about your favourite computer game. Your website has a chat room so that friends can discuss the game and share tips on how to play. What information will you collect from your friends when they register? Design a registration form to use on your website.

Online shopping

Buying online is called internet shopping or **e-commerce**. Internet shopping sites collect data when people buy things. Some of the data that shopping sites collect is personal data, including:

- bank details – so we can pay for the items
- address and other contact details – so the company can deliver the items.

Website owners must keep personal details safe. A criminal can use a person's bank details to steal money from their bank account. A criminal can use personal details such as a person's address to impersonate (pretend to be) the person and then commit crimes in their name.



Internet shops use **secure sites** to protect data. A secure site **encrypts** information that is sent over the internet. Encrypted data is coded. If a criminal steals the data they will not be able to read or use the data.

Is this site secure?

When you are browsing the web, there are two clues that tell you a website is secure.

There is a small locked padlock symbol.

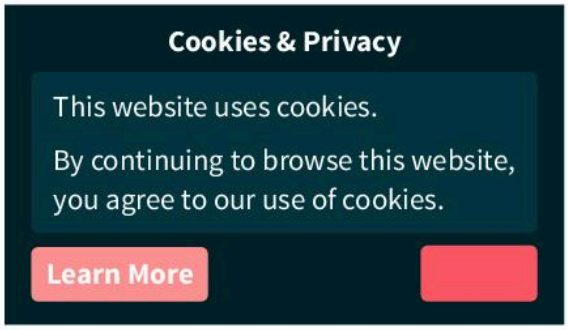


The URL of a secure site starts with https://. The 's' stands for secure. It tells you that data is encrypted when it is sent over the internet.

Never send information across the internet if the website is not secure. However, not all secure sites are safe to use. The safest way to send information is to use a secure site that you have used before and can trust.

Cookies

A **cookie** is a small file that is stored on your computer when you visit a website. A cookie saves information about the way you use web pages. The website uses this information to improve your experience of using the website.



Types of cookies

There are four types of cookie.

- **Essential cookies** are needed to make a website work as designed. On an internet shopping site, the shopping basket function needs cookies to make it work properly.
- **Performance cookies** collect information about how you use a website. The owner of a website uses this information to improve the performance of the website.
- **Functionality cookies** record what you do on a web page. This information is used to personalise the web page for you.
- **Advertising cookies** record what you look at online. The information is used to personalise the adverts displayed on a web page.

What are cookies used for?

Cookies are used for two main reasons:

- Cookies make it easier to use websites. For example, if you complete a form on a website, the information is saved in a cookie. This means that the website can automatically complete the form the next time you use it.
- Cookies are used to personalise a website. For example:
  - A cookie can store your location. That information can be used to make sure you see weather reports and events that are local to you.
  - Advertisers use cookies to make sure they show you adverts for events and products you are interested in. They can collect this information from your browsing history.
  - The news stories that you see can be customised to match your interests.



Did you know?

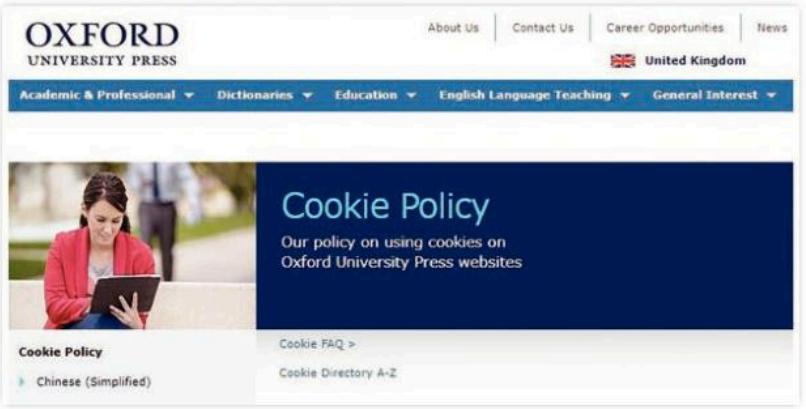
The first internet shop was called NetMarket. It made its first sale in 1994. In 2019, 1.92 billion people were expected to buy something on the internet.

Cookies and the law

The information that cookies collect is often very valuable to advertisers. Some website owners sell the information to advertisers and other organisations such as political parties. Some cookies have been developed to collect valuable information that can then be sold. These are called tracking cookies.

Some people worry that tracking cookies can be used to influence our opinions and affect the way we vote in elections.

The governments in some countries have passed laws about the use of cookies. Website owners must make it clear what cookies their sites use and what they use the cookies for. These laws make it easier for people to decide which cookies to accept and which to reject.



Activity

Draw a table with two columns like the one shown here.

Good uses of cookies	Bad uses of cookies

In the first column list all the ways that cookies make using the internet better. In the second column list all the bad things that cookies are used for.

Extra challenge

Search the web to find out more information about cookies. Add any new information you discover to the table you created in the activity.

Test

1. Name two reasons for data being collected online.
2. What are the two main reasons that websites use cookies?
3. Explain how cookies can be used to personalise a website. Give examples.
4. Why do people worry about the use of cookies on websites?

2.2 Danger online

In this lesson

You will learn:

- about the risks you face online from cybercriminals.

Spiral back

In Student Book 6 you learned how to behave responsibly online and to respect other internet users. Not every user on the internet behaves responsibly. Some users behave disrespectfully. Other users are involved in criminal activity. In this lesson and in Lesson 2.3 you will learn how to recognise criminal activity on the internet so that you can protect yourself.

Cybercrime

One risk you face on the internet is **cybercrime**. A **cybercriminal** is someone who uses the internet to commit their crimes. Some cybercriminals sell illegally over the internet. Some cybercriminals attempt to steal money and other possessions from people.

Cybercrime methods

Criminals use several methods to steal people's money and identities:

Identity theft – A criminal steals personal information such as a person's name and address. They also steal official information such as social security numbers or passport numbers. Criminals can use a stolen identity to pretend they are someone else. They can open a bank account or a credit card account in someone else's name. They then use these fake accounts to buy goods or to steal money from banks.

Phishing – A criminal creates a fake website that looks like the official website of a real bank or internet shopping website. The criminal then writes a fake email that looks like it comes from the bank or shopping website.

The email tells the person receiving the email that there is a problem with their account and they must log in to solve the problem. There is a link to the fake website. When the person logs in, the fake website records their username and password. The criminal can then use the person's login details to steal money.



Only a small percentage of people are fooled by phishing emails, but every victim can lose hundreds or thousands of dollars to the criminal.

Scamming – A criminal sends out an email that asks for money. Sometimes the email contains a threat that the victim owes money and will have to go to court if they do not pay the debt. Sometimes the email offers something for sale at a very low price or asks for a donation to a charity. Scammers also use social media sites.

If the criminal emails enough victims, someone will be fooled and will send money to the criminal.

Hacking – A criminal breaks into a computer system, usually to steal files or personal information. Hackers usually try to break into computer systems that hold a lot of personal information. Bank systems are an example of the type of system that hackers attack.



Did you know?

Banks and other organisations sometimes employ hackers to attempt to break into their systems. This can reveal weak spots where the system is insecure. The organisation can then fix the weak spots before a malicious hacker finds them.

Be creative

Create a poster that warns about the dangers of cybercrime. Use some or all of the key words from this lesson. Search the internet for a good image to illustrate your poster. Your poster can warn about all the dangers of cybercrime or it can focus on one type of cybercrime, such as identity theft.

Malware

Malware is software that a criminal installs on a computer. Some types of malware are designed to steal personal information. Other malware is designed to damage files or stop a computer from working properly. Some types of malware are:



- **Virus** – spreads and infects files that it meets. That makes a virus difficult to cure. Some viruses are relatively harmless – they make messages pop up on your screen, for example. Other viruses are more dangerous – they can destroy files and steal personal data.
- **Trojan** – malware that is hidden inside another program, for example in a game. When someone uses the program, the malware is released and starts to work. Trojans are popular with cybercriminals because they are easy to write and spread.
- **Ransomware** – encrypts all the data on a computer. That means that the data cannot be used. The victim of a ransomware attack receives a message demanding a payment to remove the encryption.
- **Spyware** – sits on a computer and records what the user is typing. Criminals can use spyware to discover login details.
- **Adware** – places unwanted adverts on a computer. The adverts are often displayed in another web page such as a social media or internet shopping site. Adware is not as damaging as other malware, but it can be annoying. If you have adware on your computer, it is a warning that your computer is vulnerable to other more dangerous malware.



Criminals often use email to spread malware. They often send fake emails with subjects that encourage you to open them. For example, the email subject might say you have won a prize in a competition or warn you that your computer has a serious problem you need to deal with.

The email encourages you to open or install a software application or open a document. The file contains malware, so when you open the file your computer is infected with the malware.



Did you know?

One of the most serious malware attacks was a ransomware attack called WannaCry. It affected over 200,000 computers in 150 countries. WannaCry closed important computer systems including some hospital systems, so the hospitals had to cancel operations. The cost of the attack was estimated at $4 to $8 million.

Activity

Read the Did you know? box. Search the internet to find out about other major malware attacks that have taken place. Choose one of the malware attacks that interests you. Write a brief description of the attack. Some things you can include in your description are:

- the name of the malware
- the type of malware
- the effect of the malware
- when the attack took place
- how many people were affected
- the cost of the attack
- the person or people responsible for the attack.

Extra challenge

Trojan malware is sometimes called 'Trojan horse' malware. Its name comes from an Ancient Greek story. Search the internet to find out what the original Trojan horse was. Why is the name Trojan horse used for this type of malware?



Test

1. Name three types of malware.
2. What is ransomware and why is it dangerous?
3. Explain why a stolen identity can be valuable to a criminal.
4. Describe how criminals use emails to spread malware.